

SARDAR PATEL UNIVERSITY, VALLABH VIDYANAGAR

EXTERNAL EXAMINATION (OLD COURSE)

BCA- SEMESTER-1

US01FBCA02- MATHEMATICS

DATE: 28-01-2021

TIME: 10:00AM TO 12:00PM

MARKS: 70

Q.1 MULTIPLE CHOICE QUESTIONS

[10]

1. Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8\}$ then $A \cup B = \dots\dots\dots$
 (A) $A \cup B = \{1, 2, 3, 4, 6, 8\}$ (B) $A \cup B = \{2, 4, 6, 8\}$
 (C) $A \cup B = \{2, 4, 6\}$ (D) $A \cup B = \{2, 3, 4, 5, 6\}$
2. If $U = \{1, 2, 3, \dots\dots\dots, 9\}$ and $A = \{1, 2, 3, 4\}$
 Then $A^c = \dots\dots\dots$
 (A) $\{1, 2, 3, 4\}$ (B) $\{5, 6, 7, 8, 9\}$ (C) $\{3, 4, 5, 6\}$ (D) $\{1, 3, 5, 9\}$
3. $A \cap A = \dots\dots\dots$
 (A) Φ (B) A (C) U (D) None
4. Let $A = \{0, 1\}$. Is A closed under multiplication?.....
 (A) Yes (B) No (C) Both A & B (D) None
5. $(a^{-1})^{-1} = \dots\dots\dots$
 (A) e^{-1} (B) a (C) e (D) none
6. $a^{-1}a = \dots\dots\dots$
 (A) A (B) e (C) A^{-1} (D) none
7. If $u = (1, 2)$ and $v = (3, 2)$ then $u + v = \dots\dots\dots$
 (a) $(4, 3)$ (b) $(4, 4)$ (c) $(3, 4)$ (d) $(1, 4)$
8. If $u = (a, b)$ and $v = (c, d)$ then $u \cdot v = \dots\dots\dots$
 (a) $(ab + cd)$ (b) $(ac + bd)$ (c) $ac + bd$ (d) $ad + bc$
9. _____ is the value dividing the data into two equal parts.
 (a) mode (b) median (c) mean (d) none
10. The arithmetic mean for simple frequency distribution is given by _____
 (a) $\frac{\sum Xi}{n}$ (b) $\frac{\sum fixi}{n}$ (c) $\frac{\sum fixi}{N}$ (d) none

Q.2 FILL IN THE BLANKS AND TRUE OR FALSE

[08]

1. $u = (1, -2, 3, 4)$, $w = (5, -5, -3, 4)$ so $u + w = \dots\dots\dots$
2. If $u = (1, 2, 3, -4)$, $v = (5, -6, 7, 8)$ so $u \cdot v = \dots\dots\dots$
3. The geometric mean for the frequency distribution is given by _____
4. The harmonic mean is given by _____
5. The norm of a vector d is denoted by $\|d\|$. True/False
6. If $u = (1, 2)$ and $v = (2, 2)$ then $u + v = (4, 3)$. True / False
7. The length of vector w is denoted by $\text{len}(w)$. True/False
8. Let $A = \{0, 1\}$. It is A closed under multiplication. True /False

Q.3 SHORT QUESTIONS (ANY TEN)

[20]

1. $X = \{\text{red, blue}\}$, $Y = \{\text{blue, green, orange}\}$, $Z = \{\text{red, blue, white}\}$
 $U = \{\text{red, yellow, blue, green, orange, purple, black, white}\}$
 Find (1) $x^c \cap (y-z)$ (2) $(x \cup y)^c$
2. Determine the power set of $A = \{a, b, c\}$

- 3 Define Complement of a set with example.
- 4 Define * on Q by $a * b = a + b - ab$. Is it associative? Justify Your answer
- 5 Define Duality . Find duality of $(X \cup Y)^c = X^c \cup Y^c$
- 6 Let $B = \{ 1, 2 \}$.Is B closed under : (a) Multiplication (b) addition
- 7 Consider $u = (1, -2, 3, 4)$, $v = (-2, 4, 5, -3)$ $w = (5, -5, -3, 4)$ and $z = (2, 7, 4, -2)$. Find the followings: (i) $u + v$ (ii) $-3u$ (iii) $-w$ (iv) $3u - 4w$
- 8 Find x and y : if $(x, 7) = (4, x + y)$
- 9 If $u = (1, 2, 3, -4)$, $v = (5, -6, 7, 8)$, $w = (1, -2, 5, 3)$
then find : $(u + v) \bullet w$
- 10 The intelligence quotients (IQ's) of 10 boys is given below:
70, 120, 110, 101, 88, 83, 95, 98, 107, 100 Find the Mean
- 11 The following data indicate the number of children of 30 families. Prepare the frequency distribution based on the data.
0 1 2 0 3 1 1 2 2 2 3 0 2 1 1 1 2 3 2 2 1 1 2 2 3 2 1 1 2 1
- 12 Prepare an inclusive frequency distribution consisting of six classes by classifying raw data of heights (in cms) of 30 students.
165 153 158 149 152 145 162 151 155 148 141 149 157 148 168 162
141 145 152 150 149 154 160 162 153 161 150 159 148 163

Q.4 WRITE LONG QUESTION ANSWER (ANY FOUR)

[32]

- 1 By using Mathematical induction Method
Prove that $1 + 3 + 5 + \dots + (2n - 1) = n^2$
- 2 Find a formula for the inverse of $f(x) = \frac{2x-3}{5x-8}$
- 3 Consider the set of rational defined by $a * b = a + b - ab$
Find $(7 * 1/2)$ Is $(Q, +)$ semigroup? Is it commutative?
- 4 Consider the set of rational and * is defined on Q by $a * b = a + b - ab$,
Is $(Q, *)$ commutative? Prove that $(Q, *)$ is monoid. Also find it's inverse if it exist
- 5 Find the transpose of a matrix

$$(i) A = \begin{pmatrix} 1 & 0 & 3 & 2 \\ 9 & -1 & 5 & 4 \\ 4 & 7 & 5 & 8 \end{pmatrix} \quad (ii) B = \begin{pmatrix} 1 & -5 & 3 & 2 \\ 4 & -1 & 4 & 4 \\ 4 & -6 & 5 & 8 \end{pmatrix}$$

$$6 \quad \text{Find the inverse of } B = \begin{pmatrix} 1 & -2 & 2 \\ 2 & -3 & 6 \\ 1 & 1 & 7 \end{pmatrix}$$

- 7 Calculate the mean, median, mode for the following data.

xi	fi
10	14
30	23
50	27
70	21
90	15

- 8 The following table gives the weights of 31 persons in a sample enquiry. Calculate mean, Median and Mode.

Weight	No. of Persons	Weight	No. of Persons
130	3	148	5
135	4	149	2
140	6	150	1
145	6	157	1
146	3		

